

# Dataset description

## 1. Admissions Table

This table is the core of the dataset, containing one row for each specific ward/clinic stay. It provides the demographic, temporal, and procedural context for every patient interaction.

- Identifiers (*patient\_id*, *hospitalization\_id*, *stay\_id*): These IDs allow you to track an individual patient across their entire history, group stays into a single hospital episode, and analyze each specific ward visit.
- Demographics (*patient\_age*, *patient\_sex*): Standard demographic features essential for risk modeling and bias analysis.
- *hospital\_admission\_procedure* (admission mode): This is a critical categorical field that directly informs whether a readmission was unplanned. Its values provide a clear signal, simplifying a major analytical challenge.
  - Key Values:
    - *emergency admission...*: Unplanned. Any readmission with this value will be counted in the numerator.
    - *planned admission based on a referral*: Planned. Readmissions with this value will be excluded.
    - *transfer from another hospital*: Can be used to identify patients with potentially more complex care paths.
- *hospital\_discharge\_procedure* (discharge mode): This field is crucial for correctly defining the eligible patient cohort (the denominator).
  - Key Values for INCLUSION (standard discharges):
    - *completion of the therapeutic or diagnostic process*: A general, successful discharge.
    - *referral for further treatment in an outpatient setting*: A successful discharge with a planned outpatient follow-up.
  - Key Values for EXCLUSION (non-standard discharges):
    - *referral for further treatment in another hospital*: These are transfers.
    - *patient's death*: These patients cannot be readmitted.
    - *discharge on own request*: The discharge was not clinically determined.

## 2. Diagnosis Table

This table provides the "why" behind each hospital stay and is the primary source for clinical features. It links to the Stays Table via *stay\_id*.

- *diagnosis\_code*: The raw ICD-10 code for a patient's condition.
- *type\_of\_diagnosis* (diagnosis type): A high-level category for the diagnosis.
  - Values: Preliminary, Clinical, Of death, Oncological. This can be used to filter or analyze specific diagnosis events.
- *kind\_of\_diagnosis* (diagnosis kind): This is an exceptionally valuable field for feature engineering. It provides a pre-classified context for each diagnosis code, saving significant effort.
  - Values & Utility:
    - Main / Primary: Use this to identify the primary reason for the hospital stay. This is essential for understanding the index admission.
    - Comorbid: This is a direct flag for a co-existing condition. You can create a powerful feature - *count\_of\_comorbidities* - by simply counting how many times this value appears for a given stay. This is a fast and effective proxy for patient complexity.
    - Secondary: Can be used alongside "Comorbid" to get a full picture of the patient's health status.

# Task 1: Unplanned Readmission Classification

## Task Definition

The first and most critical goal is to create the "ground truth" for this project. The task is to transform the raw hospital data into a clean, analytical dataset with a single, clear target variable:

***is\_unplanned\_readmission***. This label will be the foundation for all subsequent analysis and modeling.

## Proposed approach:

1. Data Preparation: The process begins by joining the Admission table with the Diagnosis table using their common key (*stay\_id*). All date columns should be converted to the correct format.
2. Defining the Cohort (Denominator): The goal is to create a clean set of "index admissions" - initial stays that are eligible for a potential readmission. The data must be filtered to include only stays that meet these criteria:

- Patient age is 18 or older.
  - The patient did not die during the stay (`did_patient_die = 0`).
  - The discharge was not a transfer to another hospital.
  - Exclude stays for psychiatry, specific medical cancer treatments, and COVID-19.
  - Exclude any admission that is *already* a readmission.
3. Identifying and Classifying Readmissions (Numerator): Now, for every eligible stay in the cohort, it's necessary to find out if it was followed by an unplanned readmission.
- Find the Next Admission: For each `patient_id`, the stays should be sorted chronologically. For each index admission, the patient's *next* admission must be found.
  - Check the 30-Day Window: The time between the `stay_end_date` of the index admission and the `stay_start_date` of the next admission must be calculated. If it's 30 days or less, it's a readmission.
  - Apply the "Unplanned" Logic: For every readmission found, it must be classified based on the rehospitalization definition. The readmission is unplanned if:
    - Its `hospital_admission_procedure` is listed as an emergency, OR
    - Its associated diagnoses in the Diagnosis table include an ICD-10 code for a "complication of surgical and medical care".
4. Creating the Final Label: Based on this logic, the final target column (`is_unplanned_readmission`) should be created on the index admission records. A 1 means the stay was followed by an unplanned readmission; a 0 means it was not.

## Task 2: Exploratory Data Analysis (EDA) & Bias Detection

### Task Definition

The goal here is to tell a compelling story with the data. With a labeled dataset, the task is to explore it to find actionable insights. The analysis should go beyond simple statistics to uncover the "who, what, when, and where" of unplanned readmissions at the hospital.

#### Guiding questions

- Who is coming back? Analyze the patient demographics. What is the readmission risk profile based on `patient_age` and `sex`? Is there a significant difference in readmission rates for patients with a high `count_of_comorbidities` (calculated from the `kind_of_diagnosis` column)?
- What kinds of stays lead to readmission? Investigate the characteristics of the initial hospital stay. Does a longer `length_of_stay` correlate with a higher or lower readmission risk? Are there specific primary diagnoses (identified via `kind_of_diagnosis = 'Main'`) that are particularly high-risk?

- Where are the hotspots? Analyze the `clinic_name` data. Do certain wards or clinics have a disproportionately high volume or rate of patients who are readmitted? This could point to areas needing more support.
- Is the system fair? This is the bias detection task. Are there noticeable disparities in readmission rates based on a patient's geographical location (using a generalized `postal_code`)?

## Task 3: Predictive Modeling

### Task Definition

The goal is to build a model that can predict the future. Using the engineered features, the task is to train a machine learning model to predict the *is\_unplanned\_readmission* label for a patient at the time of their discharge.

### Proposed approach

1. Engineering Features: A final set of features for the model should be created. This should include:
  - *length\_of\_stay*
  - *patient\_age*
  - *count\_of\_comorbidities* (from *kind\_of\_diagnosis*)
  - *number\_of\_prior\_admissions* (by counting previous stays for each patient)
  - Encoded categorical features (like *sex* and *clinic\_name*).
2. Establishing a Baseline: A simple Logistic Regression model should be built to serve as a baseline for performance and interpretability.
3. Building an Advanced Model: To capture more complex patterns, a model like LightGBM or Random Forest should be developed.
4. Validating and Interpreting:
  - Validation: Since there is no hidden test set, each team must define its own validation strategy. The presentation must clearly explain how the data was split (e.g., an 80/20 train/test split) to evaluate the model's performance. The AUC and F1-score on the team's test set should be reported.
  - Interpretation: This is the most important part. The presentation must include a feature importance plot from the best model. It should be explained what the model learned. What were the top 3-5 most predictive factors? Do they make clinical sense? How do they connect to the findings from the EDA task?

# Assessment criteria

## Task 1: Unplanned Readmission Classification

- Clarity of Methodology: How clearly did the team explain their step-by-step process for filtering the eligible patient cohort and applying the classification logic for unplanned readmissions? The jury must be confident that the team understood and correctly implemented the project's core definition.
- Handling of Data Challenges: How well did the team articulate their approach to handling data quality issues, missing values, or any edge cases they encountered during this process? The presentation should briefly cover the "data reality" and the team's solutions.

## Task 2: Exploratory Data Analysis (EDA) & Bias Detection

- Actionable Insights & Storytelling: How compelling and clear was the story the team told with the data? The focus is on the quality of the insights presented—are they non-obvious, relevant, and potentially valuable for the hospital?
- Depth of Bias Analysis: How thoroughly and clearly did the team present their investigation into potential demographic or regional biases in readmission rates?

## Task 3: Predictive Modeling

*Since there is no hidden test set, the jury will assess the modeling task based on the quality and clarity of the team's own validation process and interpretation.*

- Methodological Rigor & Validation: How clearly did the team explain their modeling process and validation strategy? The presentation must describe:
  - How they split their data (e.g., train/test split, cross-validation).
  - The performance metrics (AUC, F1-score, etc.) achieved on their own test set.
  - Justification for their choice of model and features. The soundness of this process is the primary measure of the model's quality.
- Model Interpretability and Insights: This is of equal importance. How well did the team explain *what their model learned* and what it means? The presentation must include and clearly interpret feature importance plots. The jury will look for a clear connection between the model's findings and the insights from the EDA task. The goal is to explain *why* the model is effective, not just that it is.